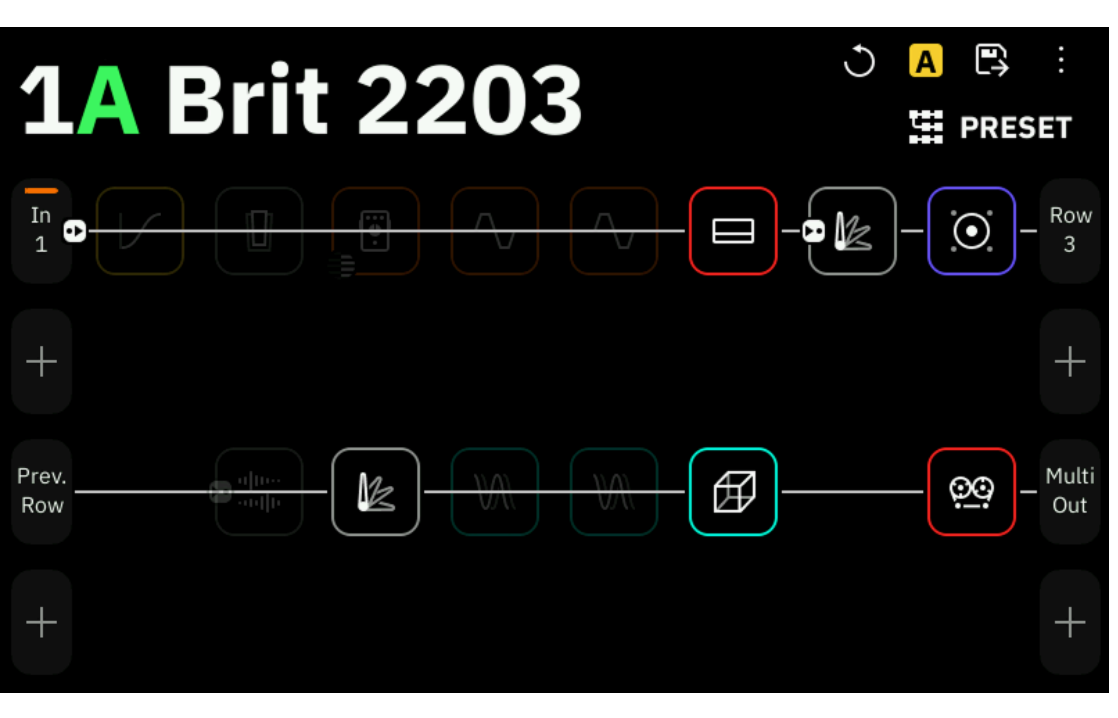
04

The Grid

Grid’s Layout

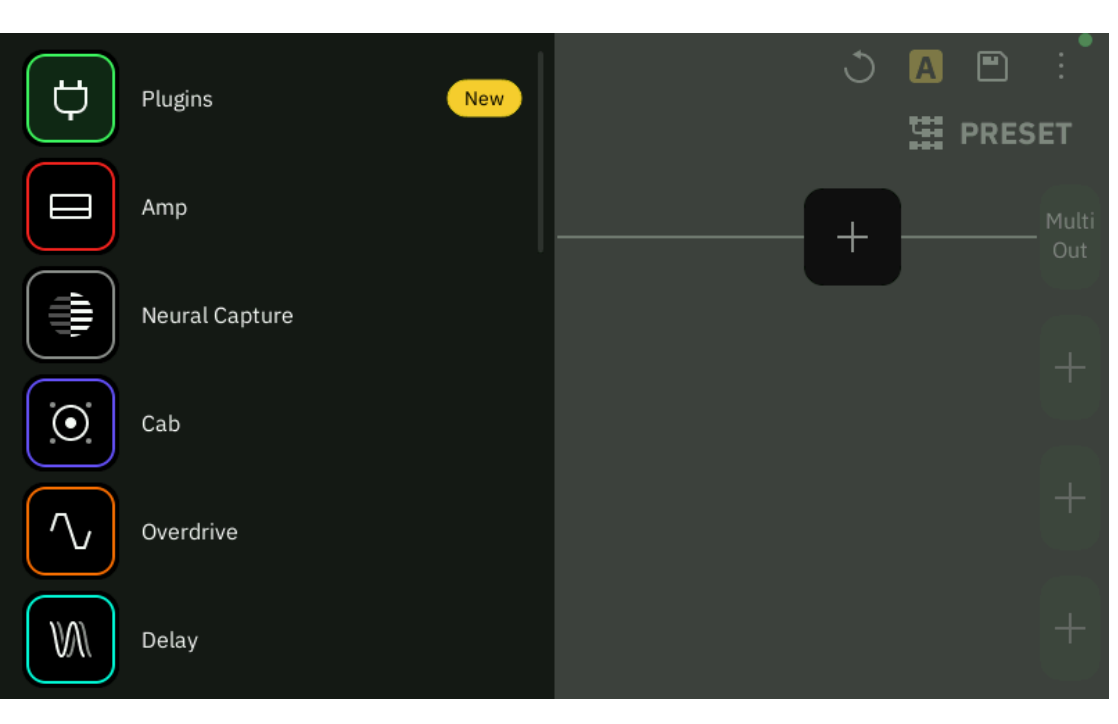


The Grid is the central workspace of Quad Cortex. It provides a clear visual layout for building Presets, offering flexibility to combine and organize device blocks as needed.

The Grid consists of **four rows**, each containing **eight device block slots**. Most elements on The Grid can be controlled using touch gestures, footswitches, expression pedals, or external devices via MIDI.

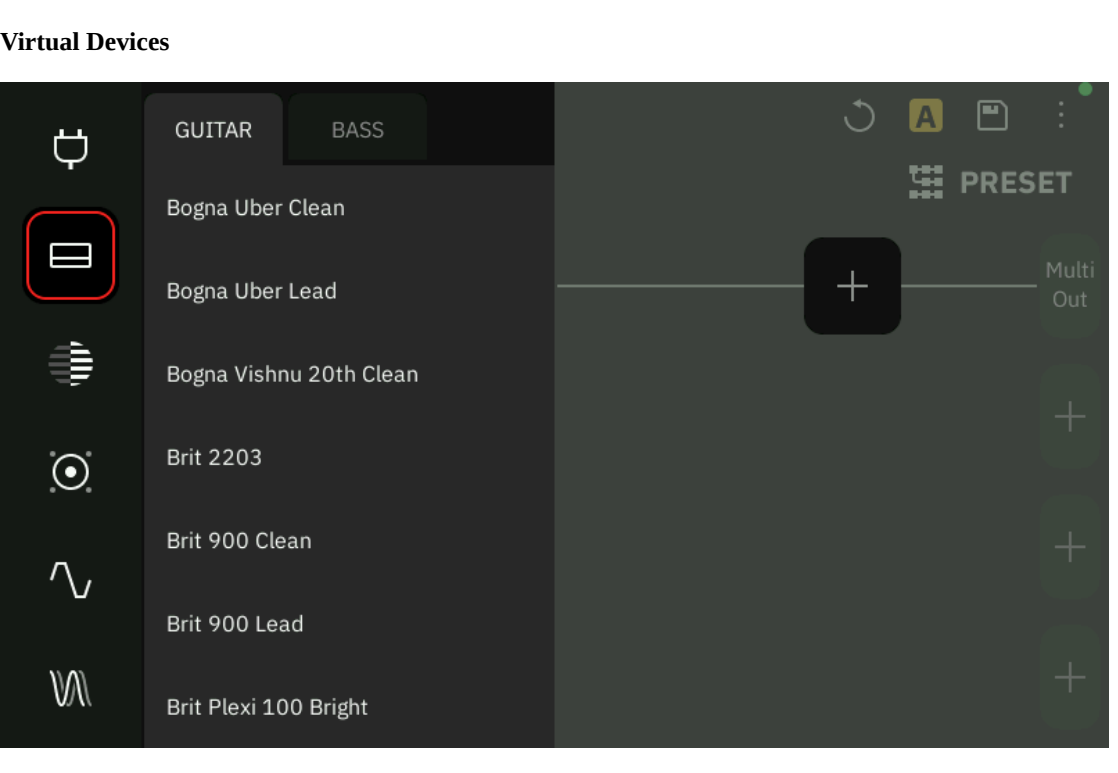
After powering on, Quad Cortex automatically displays The Grid.

VIRTUAL DEVICE LIST



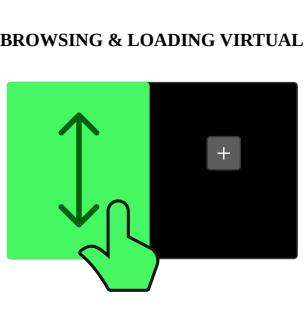
To begin building an audio signal chain, **tap an empty slot** on The Grid. This will open the Virtual Device List, which displays all device categories available.

Virtual Devices

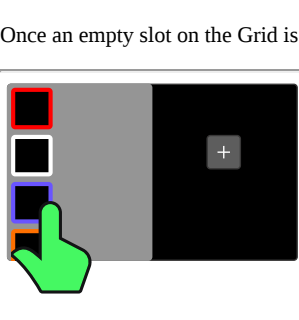


The Virtual Device List provides access to all virtual devices available that can be placed on The Grid. It allows you to browse by category, search for specific devices, and manage your favorites for quicker access.

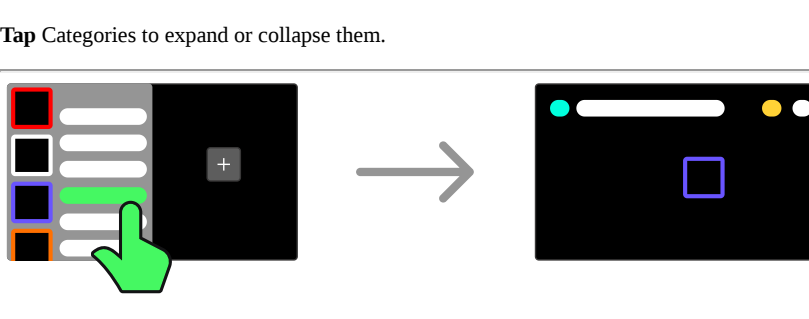
BROWSING & LOADING VIRTUAL DEVICES



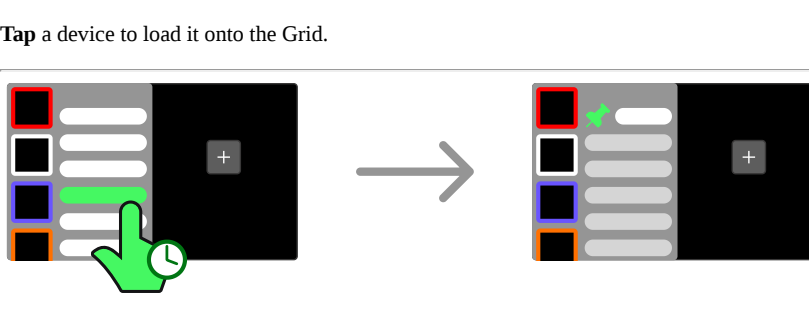
Once an empty slot on the Grid is selected, **swipe vertically** to browse virtual devices.



Tap Categories to expand or collapse them.



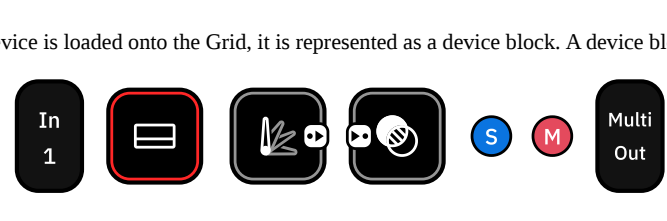
Tap a device to load it onto the Grid.



Tap and hold a device to pin it to the top of its respective category for quicker access.

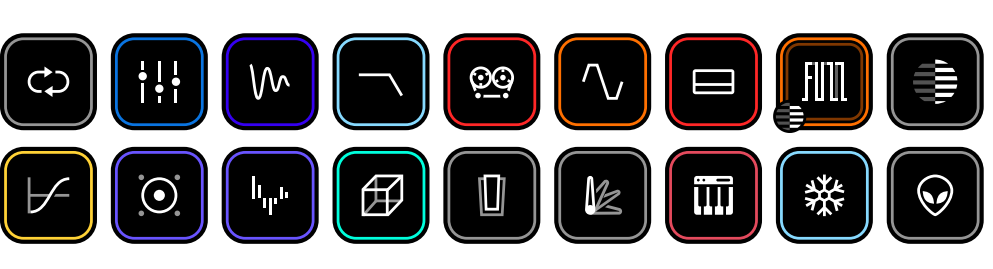
Blocks

When a virtual device is loaded onto the Grid, it is represented as a device block. A device block is a functional unit that processes audio according to the characteristics of the selected virtual device.



Device blocks can be arranged, edited, and interconnected to build signal chains within a Preset.

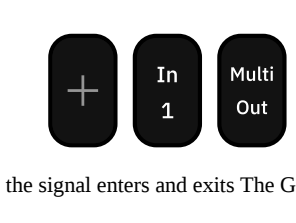
VIRTUAL DEVICE BLOCKS



Device blocks are organized into several categories, such as amplifiers, cabinets, effects, and utilities.

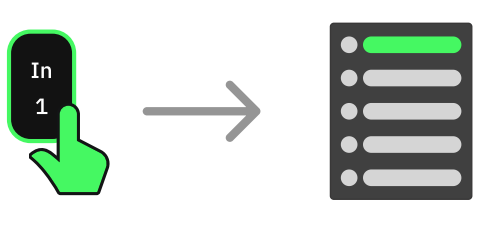
- **PLUGINS**: Compatible Neural DSP ‘X’ plugin devices.
- **AMP**: Amplifier devices for guitar and bass.
- **NEURAL CAPTURE**: Neural Capture™ devices.
- **CAB**: Cabinet simulation devices with selectable microphones, available in mono and stereo.
- **OVERDRIVE**: Boost, distortion, and overdrive pedal devices.
- **DELAY**: Digital, analog, and tape delay devices, available in mono and stereo.
- **REVERB**: Digital and analog reverb devices.
- **COMPRESSOR**: Dynamic control devices, available in mono, stereo, and side-chain.
- **PITCH**: Pitch shifter devices.
- **MODULATION**: Chorus, flanger, phaser, and other modulation devices.
- **MORPH**: Complex audio processor devices.
- **SYNTH**: Devices that generate sounds by shaping and manipulating waveforms.
- **FILTER**: Dynamic and fixed filter devices.
- **EQ**: Graphic and parametric equalizer devices.
- **IR LOADER**: Load third-party Impulse Response files.
- **WAH**: Wah pedal devices.
- **FX LOOP**: Integrate external devices to The Grid via SEND and RETURN audio ports.
- **LOOPER**: Record and layer audio in real time.
- **UTILITY**: Routing, mixing, and other audio tools.

INPUT & OUTPUT BLOCKS

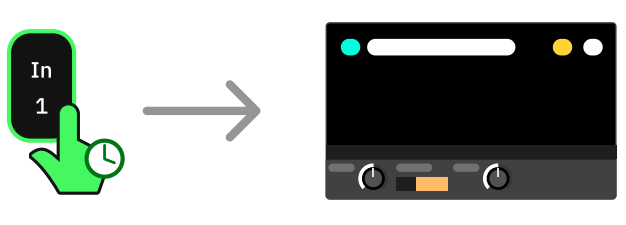


Input and output blocks define where the signal enters and exits The Grid. Their positions are fixed to ensure consistent audio routing across Presets.

- **INPUT BLOCKS**: Represent the physical inputs of the device, such as instrument inputs, microphone inputs, returns, and USB audio input channels.
- **OUTPUT BLOCKS**: Represent the physical outputs of the device, such as main outputs, sends, USB audio output channels, as well as routing to other Rows on The Grid.



Tap an input or output block to assign a different input or output source.



Tap and hold an input or output block to open its parameter editor:

- **INPUT GATE CONTROL**: Available for all assigned input blocks. Adjusts noise gate parameters and provides an additional INPUT GAIN control.
- **LANE OUTPUT CONTROL**: Available for all assigned output blocks, except when assigned to other Rows. Includes VOLUME, PAN, MUTE, and SOLO parameters.

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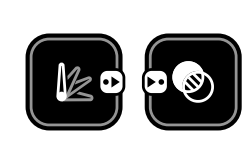
I/O Clipping Alert

The input and output blocks will turn red if their assigned I/Os are clipping.

SIDE-CHAINING DEVICE BLOCKS



Certain virtual devices support side-chain processing. Side-chaining allows a virtual device to respond to a secondary input source, rather than only the primary input determined by its position on The Grid.

Virtual devices in the **Compressor**, **Filter**, and **Utility** categories labeled **(S/C)** include side-chaining features.

- **SOURCE/TRIGGER**: Defines the secondary input used as the control signal for side-chain processing. Any input or device block that occurs earlier in the signal path on The Grid can be selected.

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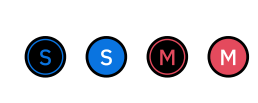
Side-Chaining Devices Limit

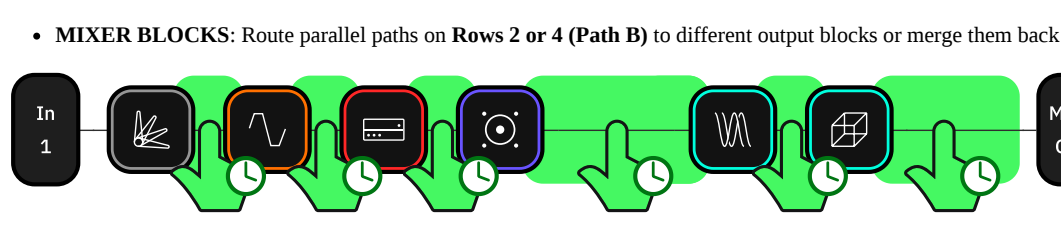
You may have up to two side-chain devices per pair of Rows within the same Preset.

SPLITTER & MIXER BLOCKS



These blocks can create parallel signal paths (A and B) and merge them back together.

- **SPLITTER BLOCKS**: Route audio from **Rows 1 or 3 (Path A)** to **Rows 2 or 4 (Path B)**, respectively, creating parallel paths that can be processed independently.
- **MIXER BLOCKS**: Route parallel paths on **Rows 2 or 4 (Path B)** to different output blocks or merge them back into **Rows 1 or 3 (Path A)**.



Tap and hold any empty slot on the Grid to insert a Splitter or Mixer for the corresponding pair of Rows. Alternatively, dragging any device block from Path A to Path B will automatically create a parallel signal path.

The **(S)** and **(M) tokens** can then be dragged and placed next to input or output blocks, as well as between device blocks.

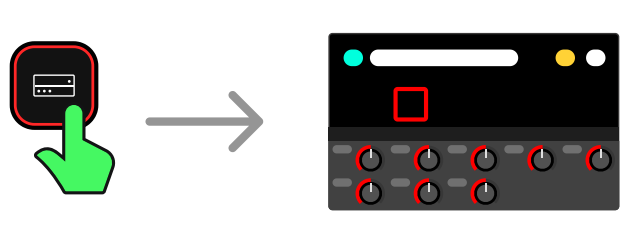
SPLITTER PARAMETERS

- **TYPE**: Determines the Splitter behavior with respect to how the audio signal is divided (Balance, A/B, or Crossover).
- **STEREO**: Enables or disables the stereo split, assigning the left and right channels of the preceding block’s output to separate paths A and B, respectively.
- **BALANCE**: Adjusts the distribution of the signal between Paths A and B (Available in ‘Balance’ type).
- **LEVEL TO A**: Adjusts the output level sent to Path A (Available in ‘A/B’ type).
- **LEVEL TO B**: Adjusts the output level sent to Path B (Available in ‘A/B’ type).
- **FREQUENCY**: Adjusts the crossover frequency at which the signal is divided (Available in ‘Crossover’ type).
- **MODE**: Swaps the crossover band assignments for Paths A and B (Available in ‘Crossover’ type).
- **MUTE**: When enabled, mutes the Splitter block.

MIXER PARAMETERS

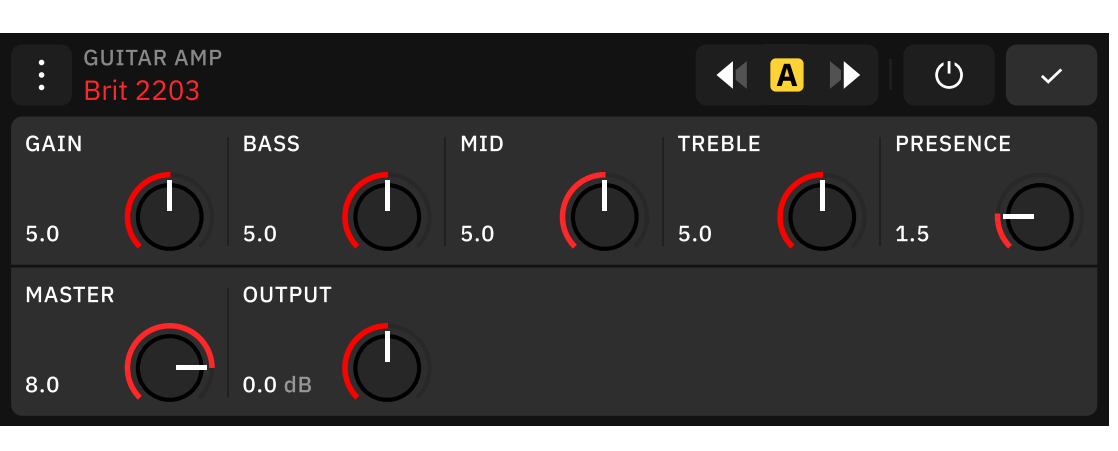
- **LEVEL A**: Adjusts the volume level of Path A.
- **PAN A**: Adjusts the stereo panning of Path A.
- **LEVEL B**: Adjusts the volume level of Path B.
- **PAN B**: Adjusts the stereo panning of Path B.
- **PHASE**: When enabled, inverts the phase of Path B.
- **MIXER LEVEL**: Adjusts the overall volume level of the Mixer block.
- **MUTE**: When enabled, mutes the Mixer block.

Parameter Editors



Parameter editors are custom layouts that appear when you **tap any block** on The Grid. They provide access to all adjustable parameters of the selected device, along with routing options, Stomp and Scene assignments, bypass state, and other advanced settings.

PARAMETER EDITOR LAYOUT



GUITAR AMP
Brit 2203

GAIN 5.0
BASS 5.0
MID 5.0
TREBLE 5.0
PRESENCE 1.5
MASTER 8.0
OUTPUT 0.0 dB

The layout of a parameter editor depends on the selected block. For most devices—including splitters, mixers, and I/O blocks—the editor occupies the bottom half of the screen. Certain devices, such as Cabs, Looper X, and other special blocks, utilize a full-screen layout to provide additional controls and options.

- **CONTEXTUAL MENU**: Provides access to additional options and device-specific actions.
- **VIRTUAL DEVICE NAME**: Displays the name of the selected device block. Tap to browse and audition other virtual devices, and replace the current device if desired.
- **FOOTSWITCH ASSIGNMENT**: Displays the current footswitch assignment. Tap to reassign or remove a footswitch assignment from the selected device.
- **SCENE SELECTOR**: Tap to navigate Scenes without closing the parameter editor.
- **BYPASS SWITCH**: Tap to bypass or activate the selected device.
- **CLOSE**: Tap to close the parameter editor and return to The Grid.
- **KNOBS**: Represent adjustable continuous parameters. Drag vertically to change their values, or tap the numeric display to enter a custom value using the on-screen keyboard.
- **DROPDOWNS**: Tap to expand a predefined set of options.
- **SWITCHES**: Tap to toggle between two or more discrete states.
- **SPECIAL PARAMETERS**: Device-specific parameters that may include advanced functions unique to certain devices (Looper X, Cabs, and other devices).

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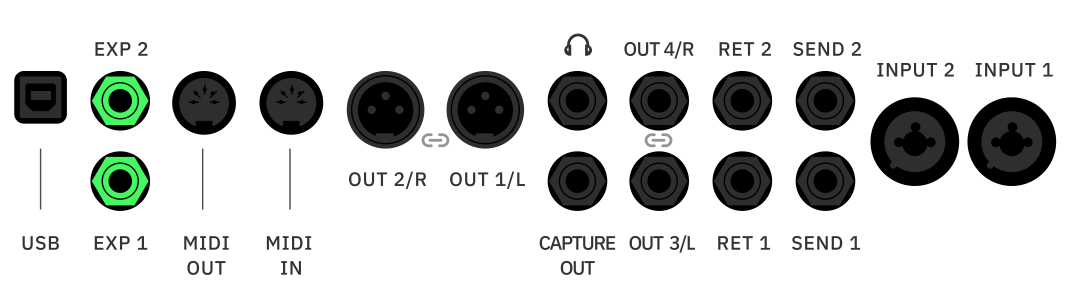
Virtual Device Cycling

When a virtual device parameter editor is open, turn **BANK UP** to cycle devices within the same category.

Expression Pedals

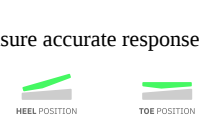
An expression pedal is a continuous controller that allows you to manipulate block parameters in real time using your foot. Common uses include controlling volume, wah, pitch, modulation depth, or any other supported parameter.



Connect expression pedals to the Quad Cortex **EXP 1** or **EXP 2** inputs using **TRS cables**.

EXPRESSION PEDAL CALIBRATION

To ensure accurate response across the full sweep range, it may be necessary to calibrate an expression pedal during first use.



1. Place the expression pedal on the surface where it will be used.
2. Connect the expression pedal to the Quad Cortex **EXP 1** or **EXP 2** inputs using a **TRS cable**.
3. Ensure the pedal is sitting flat on the surface, then access the **I/O Settings** menu (When in The Grid, swipe down from the top of the screen).
4. Tap the **EXP 1** input currently in use.
5. Tap **RECALIBRATE**, then move the expression pedal fully from heel to toe.
6. Verify that the **POSITION** indicator responds correctly, sweeping from **0.00** to **1.00**.
7. Tap **SAVE** to complete the calibration procedure.

The expression pedal calibration is stored as a global setting, remaining consistent across Presets, Scenes, and power cycles.

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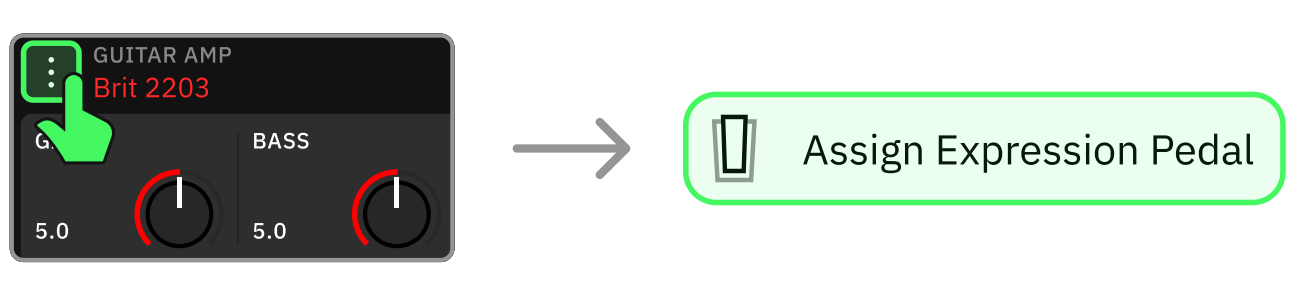
When to recalibrate?

Recalibration is recommended whenever you connect a different expression pedal model to ensure accurate operation.

EXPRESSION PEDAL ASSIGNMENT



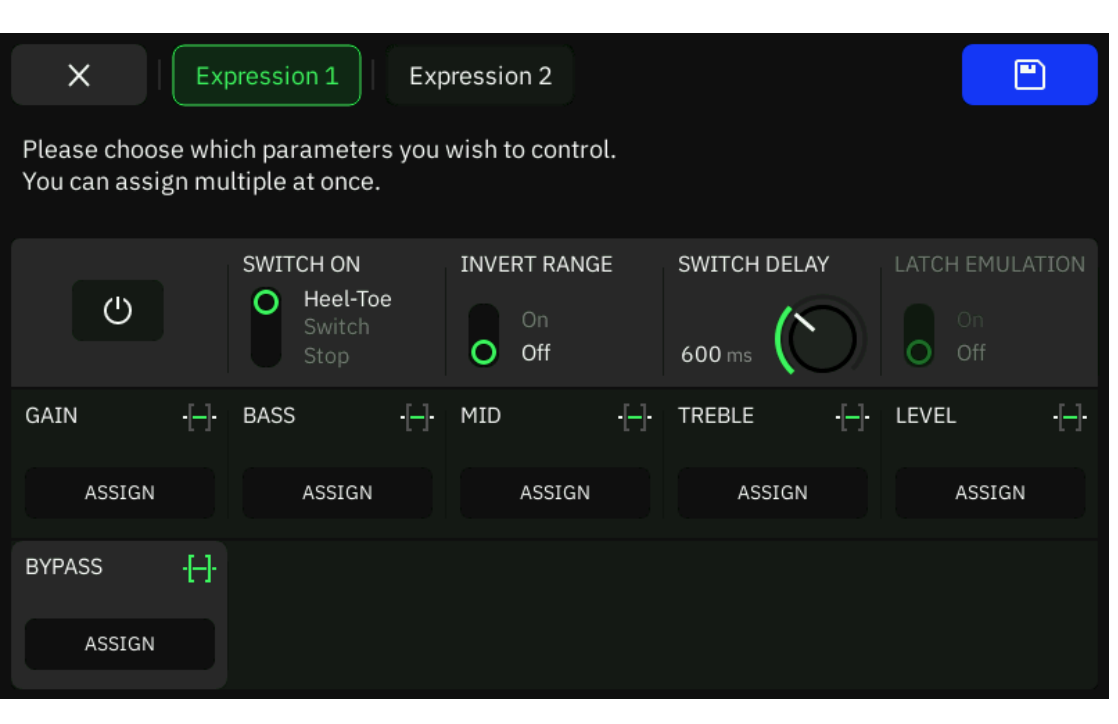
Expression pedals can be assigned to control most adjustable parameters of any block.

1. Tap a block on The Grid, access its contextual menu, and select **Assign Expression Pedal**.
2. Tap **ASSIGN** to link parameters to the expression pedal.
3. Set the **MIN RANGE** and **MAX RANGE** values for the pedal’s sweep. To reverse the behavior of a parameter, set its minimum value to 100% and its maximum value to 0%.
4. Tap **Expression 1** or **Expression 2** at the top to toggle between expression inputs.
5. Tap the **SAVE** icon in the top-right corner to confirm changes and exit.

Expression pedal assignments can be retained by using the **Set Parameters as Defaults** option in the block’s contextual menu:

- The current parameter position and toe-switch polarity are not stored when saving Presets. Instead, the current position and toe-switch state are preserved when loading Presets.
- When a parameter is assigned to an expression pedal, its values are excluded from Scene data and will not change when switching Scenes.

EXPRESSION BYPASS



Please choose which parameters you wish to control.
You can assign multiple at once.

SWITCH ON: Heel-Toe / Switch / Stop
INVERT RANGE: On / Off
SWITCH DELAY: 600 ms
LATCH EMULATION: On / Off
GAIN
BASS
MID
TREBLE
LEVEL
BYPASS

Blocks can be engaged or bypassed using expression pedals. To configure this, first assign an expression pedal to a device block by opening its contextual menu and selecting **Assign Expression Pedal**. Then, tap **ASSIGN** under the **BYPASS** parameter to access additional options:

- **SWITCH ON**: Toggles between three bypass modes (Heel-Toe, Switch, and Stop).
- **HEEL-TOE BYPASS MODE**: The block is bypassed at the heel position. If INVERT RANGE is enabled, the block will instead be bypassed at the toe position.
- **SWITCH BYPASS MODE**: Pressing the toe switch on the expression pedal bypasses the block.
- **STOP BYPASS MODE**: Holding the expression pedal still bypasses the block.
- **INVERT RANGE**: Reverses the value at which the bypass is engaged.
- **SWITCH DELAY**: Adjusts the time it takes for the block to be bypassed (Up to 5000 ms).
- **LATCH EMULATION**: Enables momentary toe switches to emulate a latching response.

EXPRESSION PEDAL COMPATIBILITY

Quad Cortex is compatible with a wide range of expression pedals. Devices with the following features are supported:

- Single TRS expression output (Wah or Volume outputs are not supported).
- Dual TRS outputs for sweep control and toe-switch.
- Latching toe switch.
- Momentary toe-switch (only supported when Latching Emulation is set to ‘On’).
- Linear or logarithmic curve.
- Minimum value knob.
- Polarity/Reverse function.

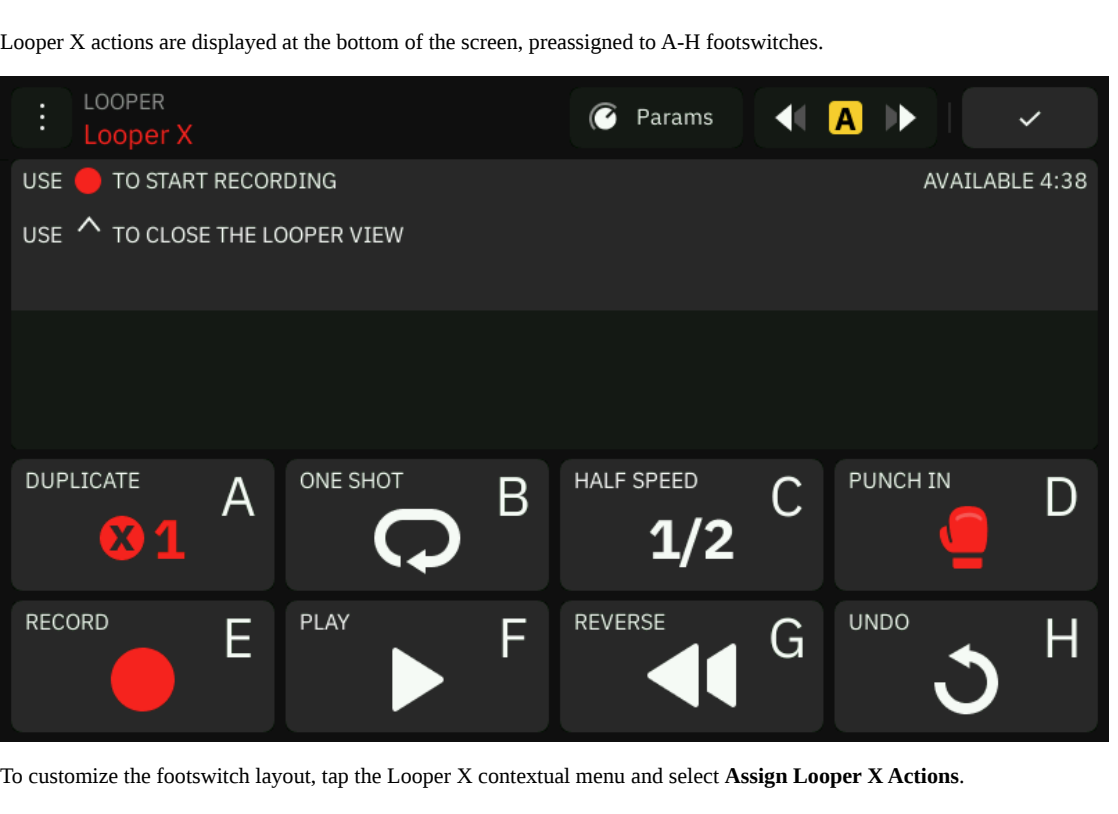
Looper X

The Looper X is a device block that, when placed on The Grid, captures and layers audio in real time, giving you full control over recording, overdubbing, playback, and more. While its position on The Grid determines whether the entire signal chain or only specific sections are recorded, custom routing options are also available.

Tap the Looper X block to open its parameter editor.

LOOPER X ACTIONS

Looper X actions are displayed at the bottom of the screen, preassigned to A-H footswitches.



To customize the footswitch layout, tap the Looper X contextual menu and select **Assign Looper X Actions**.

- **DUPLICATE**: Creates an overdub that is twice the length of the original loop, allowing you to extend the loop during overdubbing. If THRESHOLD is enabled, this action also arms the Looper for recording.
- **RE-LOOP**: Trims the length of the loop. This function is only available when DUPLICATE is active.
- **ONE SHOT**: Plays the loop once, then stops. If activated during playback, the loop stops automatically after the recorded audio ends.
- **HALF SPEED**: Plays the loop back at half the original speed.
- **PUNCH IN**: Replaces the current loop audio with a new recording. Tap again to stop recording (PUNCH OUT).
- **RECORD**: Starts recording a new loop. If THRESHOLD is enabled, this also arms the looper for recording. During playback, this control changes to OVERDUB.
- **OVERDUB**: Records additional audio on top of the existing loop during playback.
- **PLAY/STOP**: Starts or stops loop playback.
- **REVERSE**: Reverses loop playback. If activated before playback begins, the loop will start in reverse after pressing PLAY.
- **UNDO**: Reverts the last recording or overdub action. After using UNDO, the control changes to REDO.
- **REDO**: Restores any actions previously undone with UNDO.

The Looper X actions and parameters can also be controlled via **MIDI CC** messages.

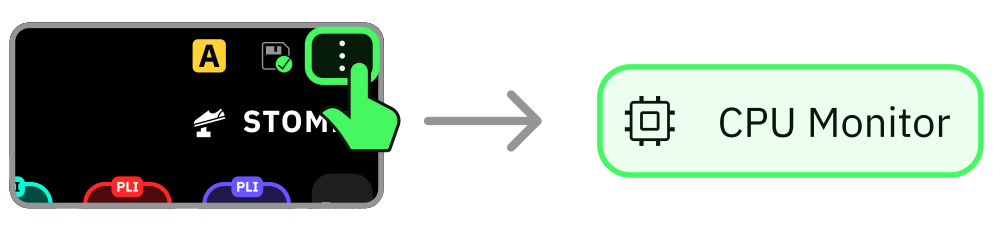
LOOPER X PARAMETERS

The Looper X parameters allow you to fine-tune playback, overdubbing, and overall loop behavior.

- **PLAYBACK LEVEL**: Adjusts the overall playback volume of the loop playback.
- **OVERDUB LEVEL**: Sets the playback level of the loop while overdubbing. For instance, a value of -5dB will reduce the volume by 5dB with each pass, gradually fading repeated overdubs.
- **HIGH PASS**: Allows high frequencies to pass while cutting low frequencies from the loop playback.
- **LOW PASS**: Allows low frequencies to pass while cutting high frequencies from the loop playback.
- **THRESHOLD**: Enables automatic recording when an incoming audio signal is detected. This function is disabled when your device is receiving MIDI clock.
- **RECORD MODE**: Sets the behavior of the RECORD function to Toggle or Momentary. In Momentary mode, recording is active only while its assigned Footswitch is held down.
- **OVERDUB MODE**: Sets the behavior of the OVERDUB function to Toggle or Momentary. In Momentary mode, overdubbing is active only while its assigned footswitch is held down.
- **DUPLICATE MODE**: Determines whether the DUPLICATE function is synced to the current tempo. When synced, loop length is locked to the tempo grid for rhythmic consistency. When unsynced, loop length is determined freely by user input.
- **PUNCH MODE**: Sets the behavior of the PUNCH IN/OUT function to Toggle or Momentary. In Momentary mode, recording is active only while its assigned footswitch is held down.
- **ROUTING MODE**: Configures the Input and Output routing for the Looper X block. The selected I/Os determine the audio path feeding the looper, allowing it to operate either as a Grid-based looper or a Global I/O looper.
- **QUANTIZE**: Syncs the loop to the current tempo or an external MIDI Clock, based on the selected number of beats per bar.
- **MIDI CLOCK START**: When enabled, receiving MIDI Clock messages automatically triggers the RECORD, DUPLICATE, and PLAY functions. This overrides the THRESHOLD recording feature.
- **PRE ROLL**: Enables a metronome count-in of 1, 2, or 4 bars before transitioning into the next state, such as RECORD or PLAY. When enabled, a visual cue follows the metronome playback count-in before transitioning to RECORD or PLAY. The upper row of white squares reflects the number of beats per bar, based on the current metronome settings. The lower row of green squares represents the number of PRE ROLL bars selected.
- **METRONOME PLAYBACK**: Toggles the metronome playback on or off.
- **RECORDING LENGTH**: Sets a fixed loop recording length, adjustable from 1 to 32 bars.

CPU Monitor

The CPU Monitor displays the overall CPU usage of the current active Preset in the top-right corner of The Grid.



To open it, tap the Grid’s Contextual Menu at the top-right corner and select **CPU Monitor**.

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Global EQ & Input Gate Bypass

The GLOBAL EQ and INPUT GATE blocks are automatically disabled when a Preset exceeds available processing resources. Notice that bypassing or disabling blocks does not reduce CPU consumption.